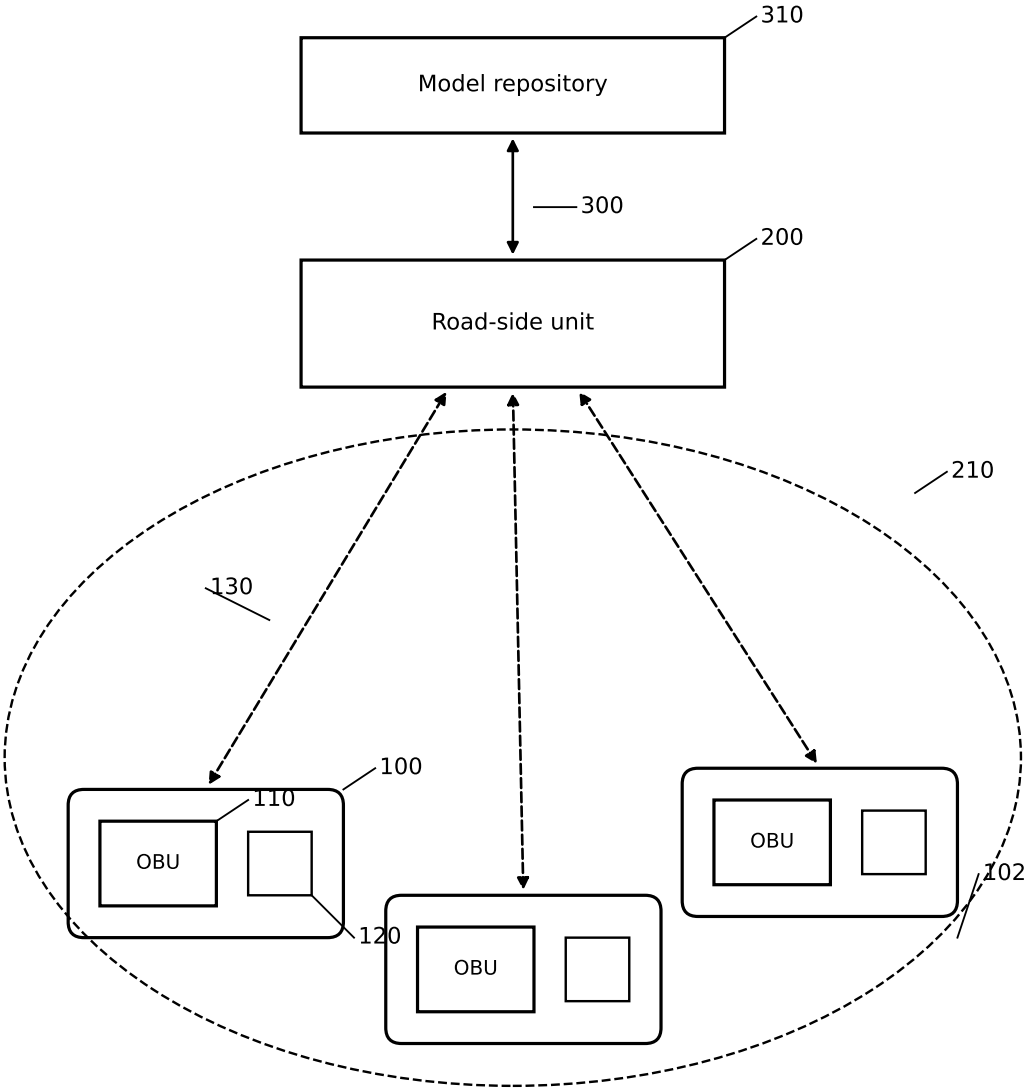

310
Model repository
300
200
Road-side unit
210
130
100
110
OBU
120
OBU
OBU
102
only masked model updates cross link 130;
raw sensor data stays in 110

FIG. 1

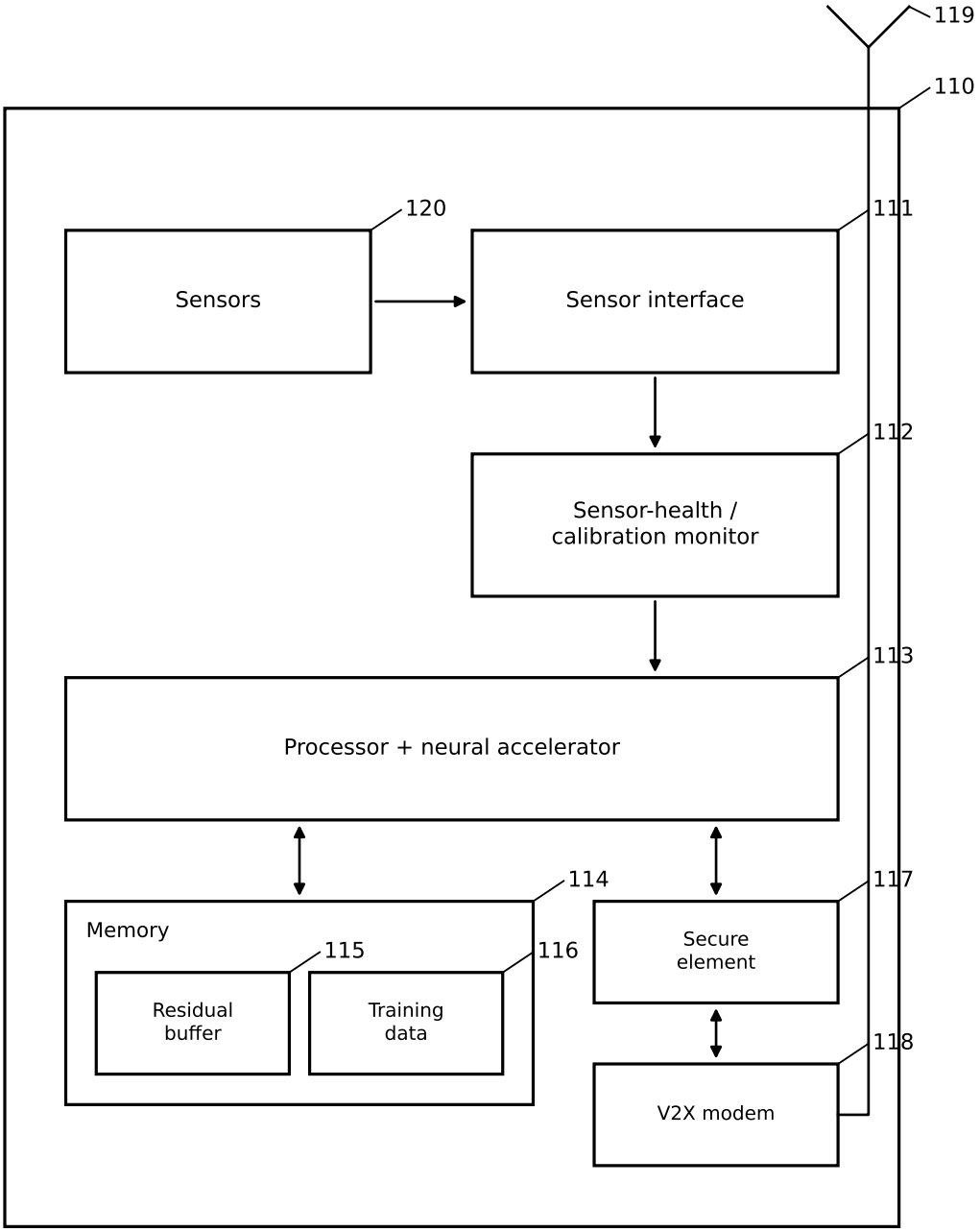

119
110
120
111
Sensors
Sensor interface
112
Sensor-health /
calibration monitor
113
Processor + neural accelerator
114
117
Memory
115
116
Residual
buffer
Training
data
Secure
element
118
V2X modem

FIG. 2

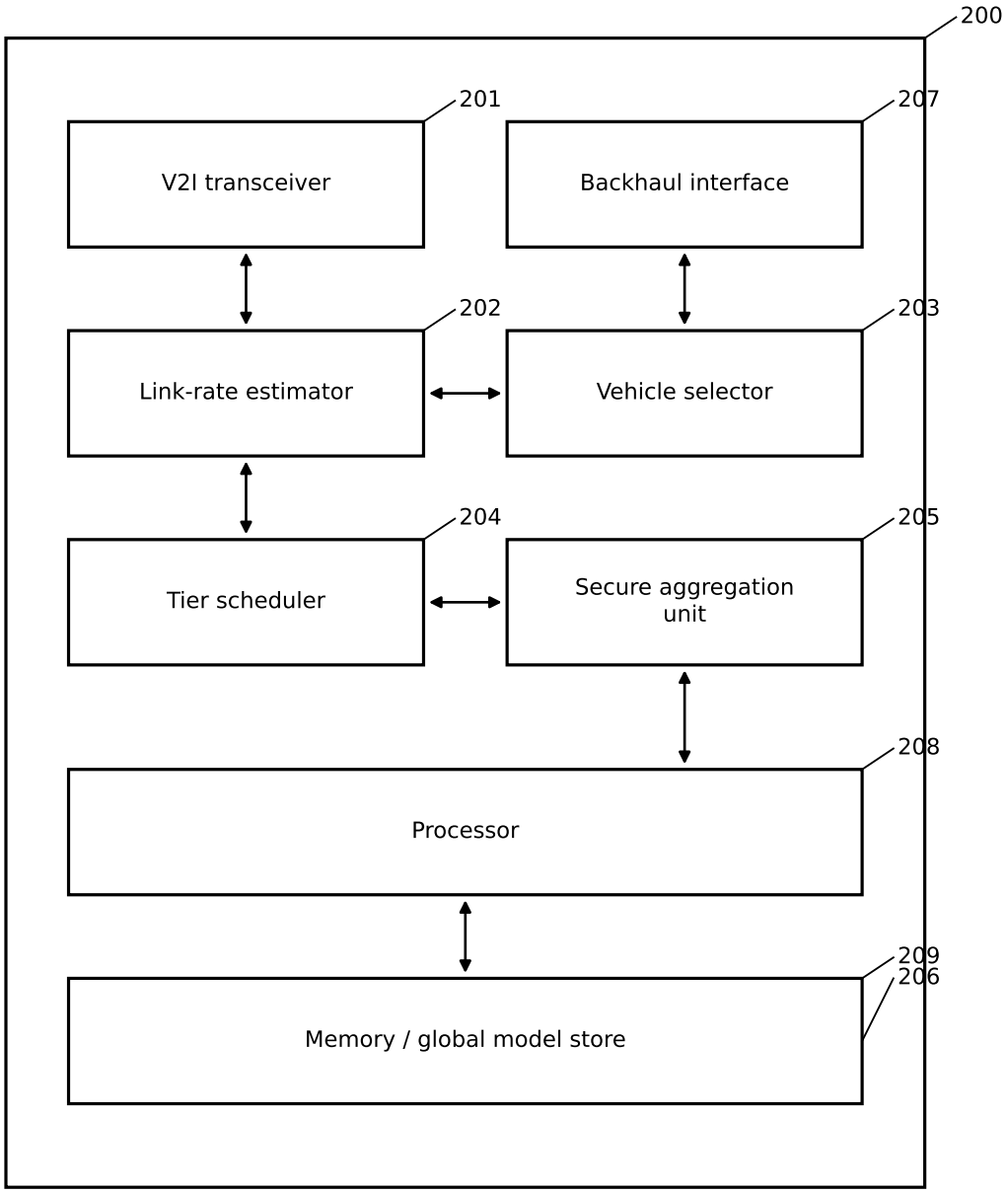

200
201
V2I transceiver
207
Backhaul interface
202
Link-rate estimator
203
Vehicle selector
204
Tier scheduler
205
Secure aggregation unit
208
Processor
209
206
Memory / global model store

FIG. 3

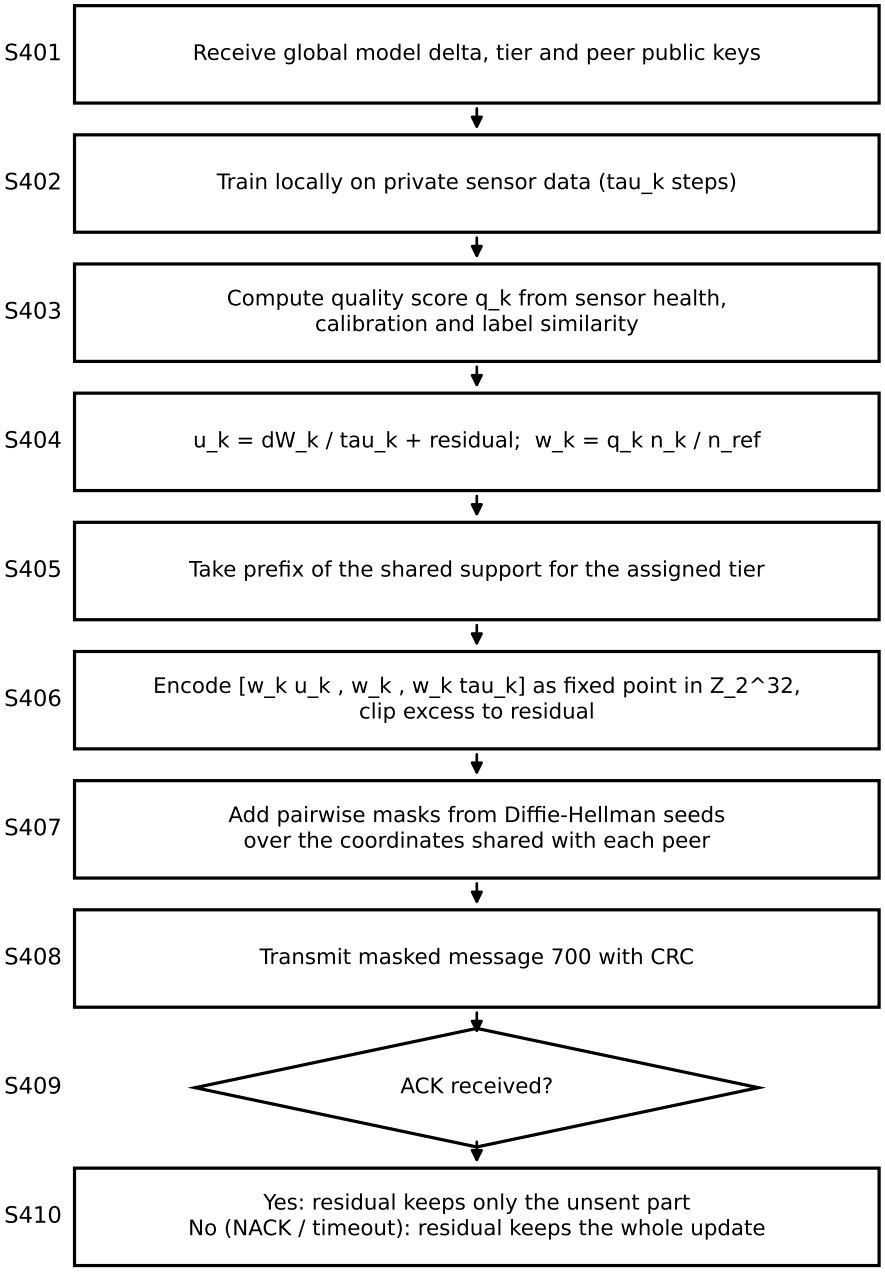

S401
Receive global model delta, tier and peer public keys
S402
Train locally on private sensor data (tau_k steps)
S403
Compute quality score q_k from sensor health, calibration and label similarity
S404
u_k = dW_k / tau_k + residual; w_k = q_k n_k / n_ref
S405
Take prefix of the shared support for the assigned tier
S406
Encode [w_k u_k , w_k , w_k tau_k] as fixed point in Z_2^32, clip excess to residual
S407
Add pairwise masks from Diffie-Hellman seeds over the coordinates shared with each peer
S408
Transmit masked message 700 with CRC
S409
ACK received?
S410
Yes: residual keeps only the unsent part
No (NACK / timeout): residual keeps the whole update

FIG. 4

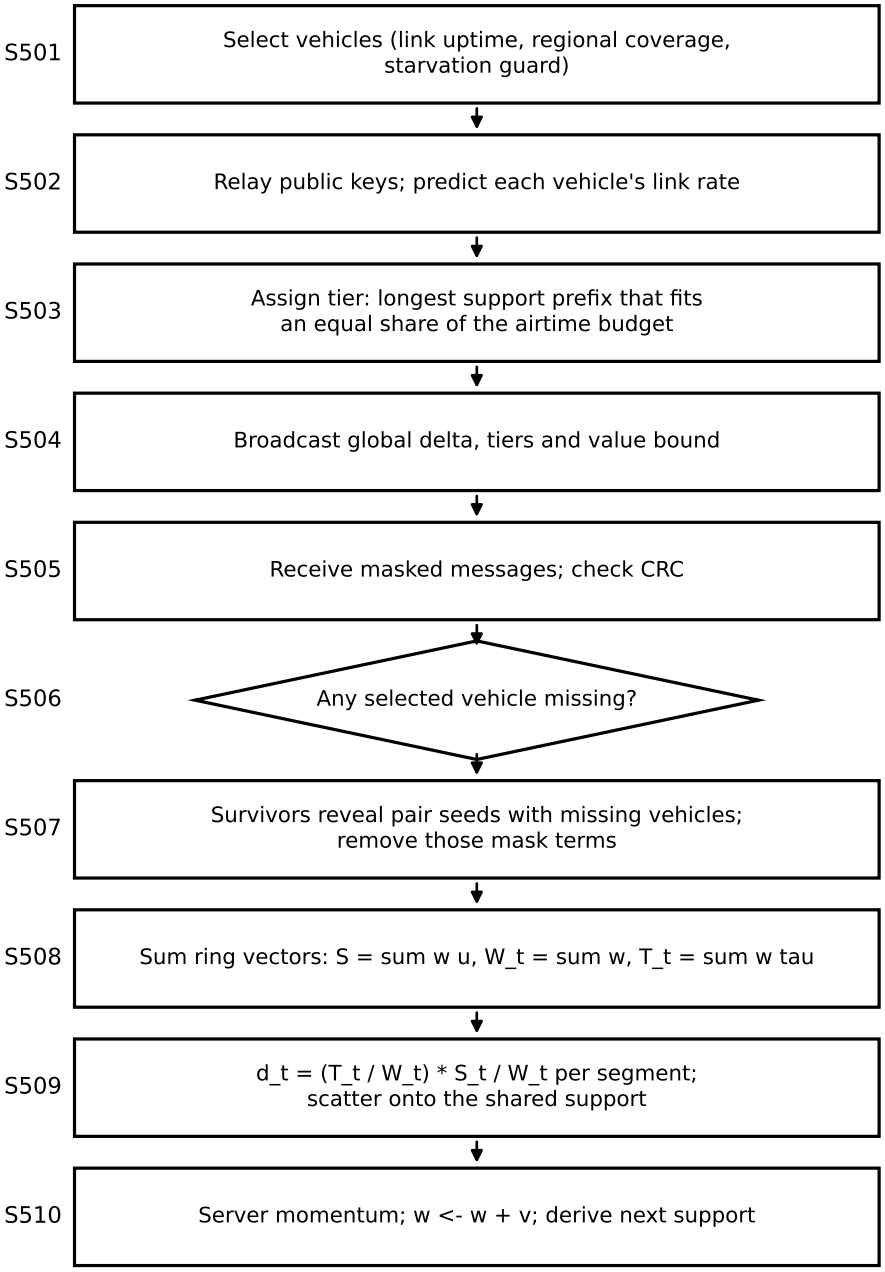

S501
Select vehicles (link uptime, regional coverage, starvation guard)
S502
Relay public keys; predict each vehicle's link rate
S503
Assign tier: longest support prefix that fits an equal share of the airtime budget
S504
Broadcast global delta, tiers and value bound
S505
Receive masked messages; check CRC
S506
Any selected vehicle missing?
S507
Survivors reveal pair seeds with missing vehicles; remove those mask terms
S508
Sum ring vectors: S = sum w u, W_t = sum w, T_t = sum w tau
S509
d_t = (T_t / W_t) * S_t / W_t per segment; scatter onto the shared support
S510
Server momentum; w <- w + v; derive next support

FIG. 5

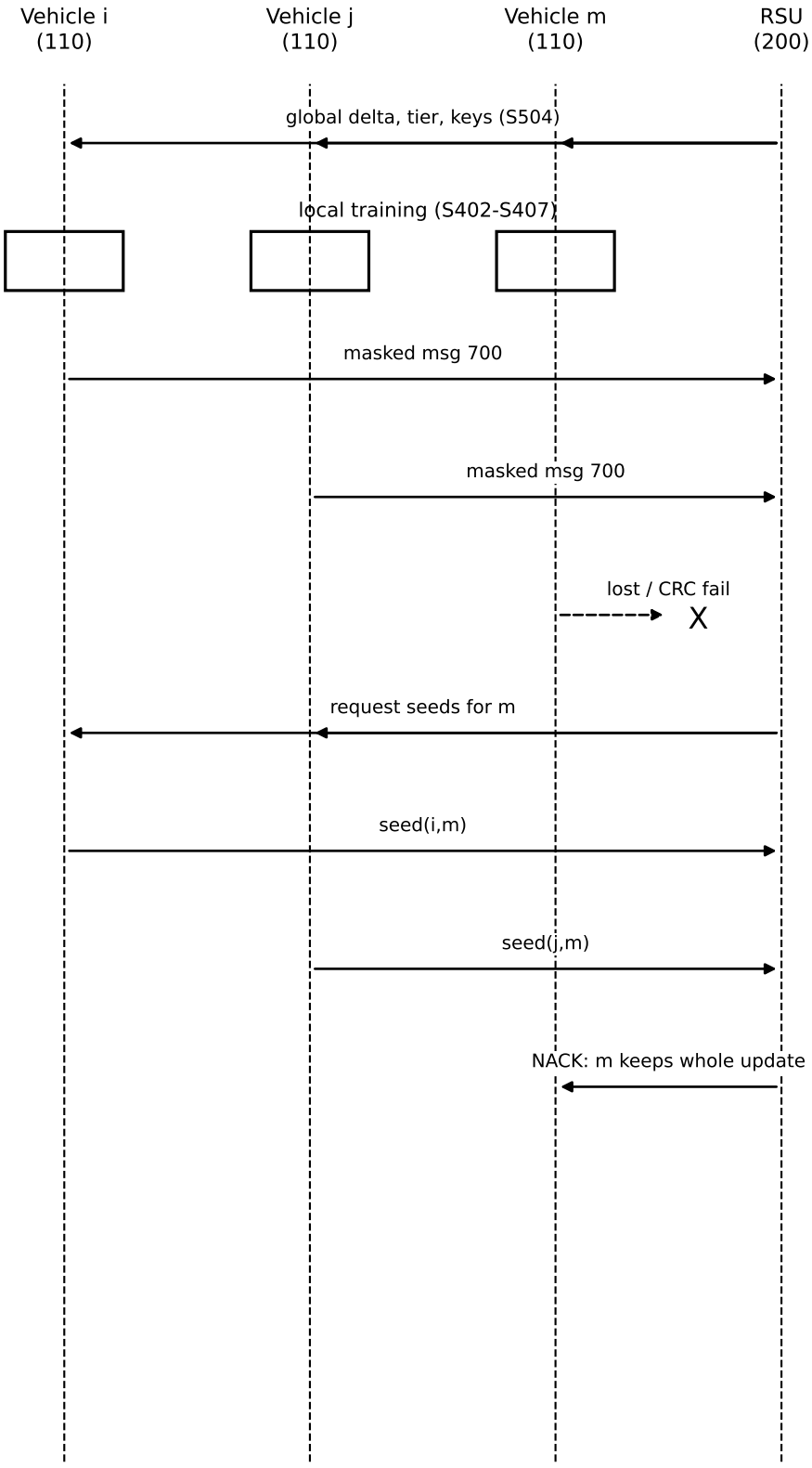
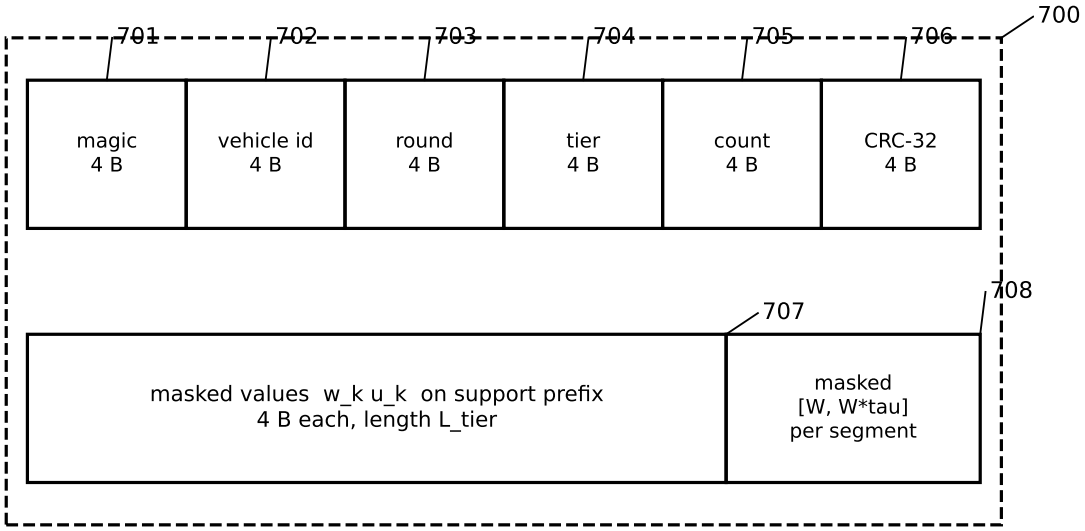
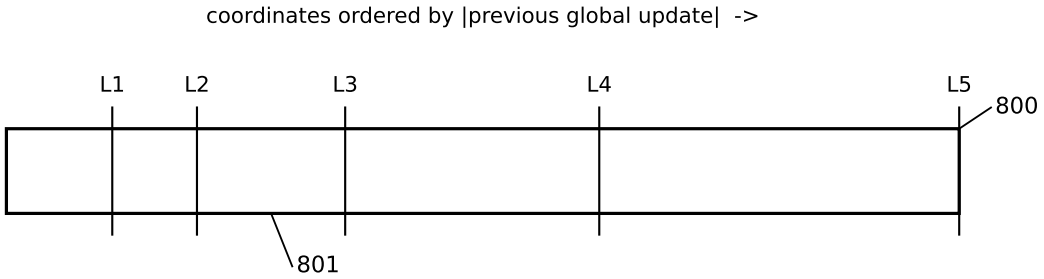

Vehicle i
(110)
Vehicle j
(110)
Vehicle m
(110)
RSU
(200)
global delta, tier, keys (S504)
local training (S402-S407)
masked msg 700
masked msg 700
lost / CRC fail
X
request seeds for m
seed(i,m)
seed(j,m)
NACK: m keeps whole update

FIG. 6



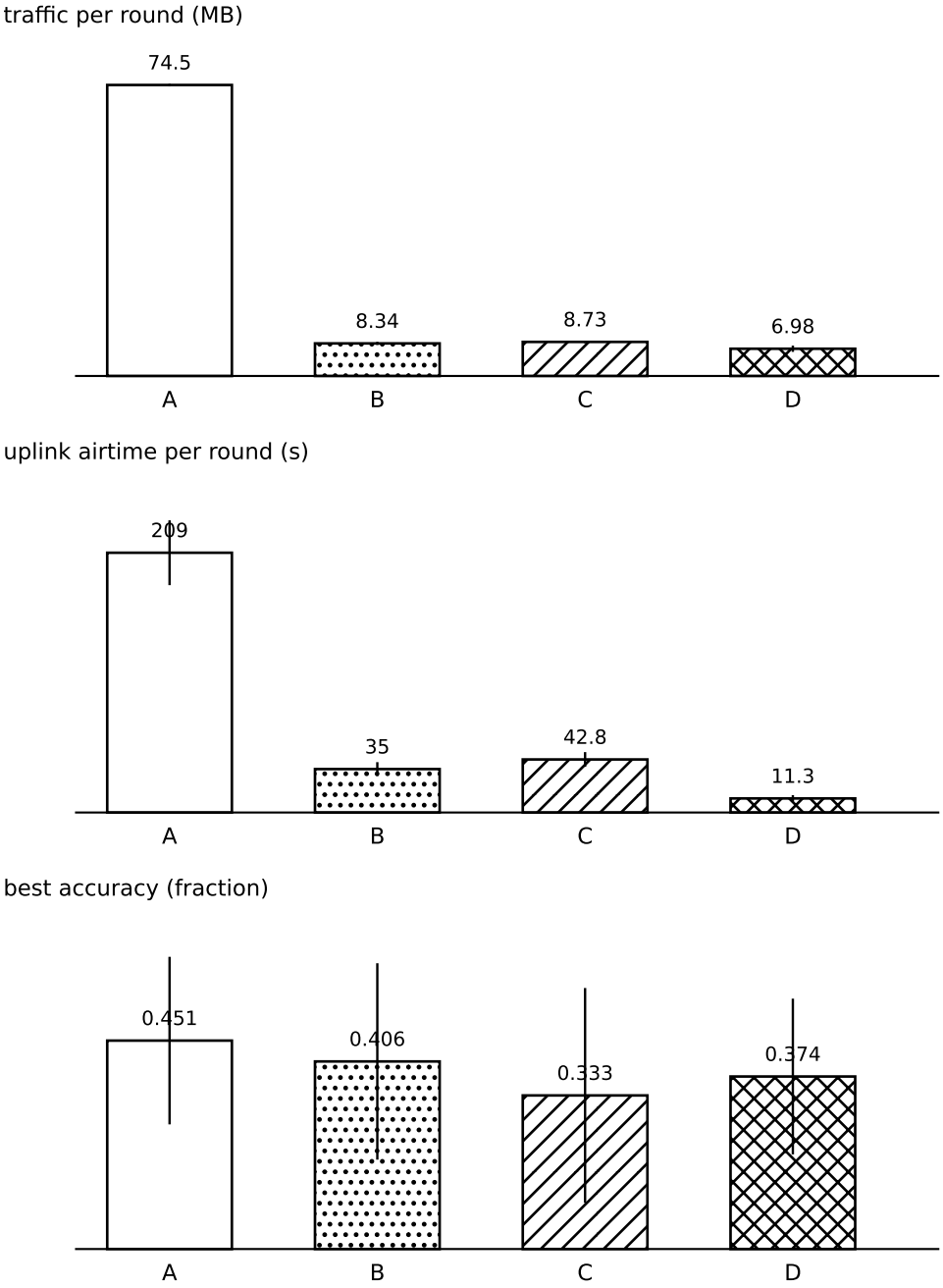
No coordinate indices are sent: every vehicle derives the same ordered support 800 from the previous global update. q_k, n_k and tau_k appear only inside masked sums.

FIG. 7



Segment t is normalised by the weights of exactly the vehicles whose prefix covers it; a pair of vehicles masks only the coordinates both of them send, so all masks cancel.

FIG. 8



A: standard FedAvg B: FedAvg on the same compression transport
C: quality-hiding secure aggregation D: C with airtime-budget tiers
bars: mean over seeds; vertical line: 95% confidence interval

FIG. 9